

Data Science Aptitude Test

Most Important Questions & Answers

Quantitative Aptitude | Logical Reasoning | Data Interpretation | Probability & Statistics | SQL & Technical Aptitude

Curated for fresher data science interview & pre-placement aptitude rounds

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Each question includes a worked answer and a quick tip. Focus on understanding the method, not just memorizing the answer — most aptitude rounds change the numbers but reuse the same logic.

1. Quantitative Aptitude

Core arithmetic and problem-solving speed — appears in almost every pre-placement aptitude test.

Q1. A shopkeeper marks an item 40% above cost price and then gives a 25% discount. What is his overall profit or loss percentage?

Answer: Let CP = 100. Marked Price = 140. After 25% discount: SP = $140 \times 0.75 = 105$. Profit = $105 - 100 = 5$, so Profit % = 5%.

Tip: Always assume CP = 100 for quick percentage-chain problems.

Q2. The average of 5 numbers is 27. If one number is excluded, the average becomes 25. Find the excluded number.

Answer: Sum of 5 numbers = $5 \times 27 = 135$. Sum of remaining 4 numbers = $4 \times 25 = 100$. Excluded number = $135 - 100 = 35$.

Tip: Average problems almost always reduce to $\text{Sum} = \text{Average} \times \text{Count}$.

Q3. A train 150 m long crosses a platform 250 m long in 20 seconds. Find the speed of the train in km/hr.

Answer: Total distance = $150 + 250 = 400$ m in 20 s $\rightarrow$ Speed = $400/20 = 20$ m/s. Converting: $20 \times (18/5) = 72$ km/hr.

Tip: Remember the conversion factor: $1 \text{ m/s} = 18/5 \text{ km/hr}$.

Q4. In how many ways can the letters of the word 'DATA' be arranged?

Answer: DATA has 4 letters where 'A' repeats twice. Total arrangements = $4!/2! = 24/2 = 12$.

Tip: For repeated letters, divide factorial of total letters by factorial of each repeat count.

Q5. If the ratio of two numbers is 3:5 and their sum is 96, find the numbers.

Answer: Let numbers be $3x$ and $5x$. $3x + 5x = 96 \rightarrow 8x = 96 \rightarrow x = 12$. Numbers are 36 and 60.

Tip: Convert ratios into a single variable (x) to solve quickly.

Q6. A can complete a work in 12 days and B can complete it in 18 days. In how many days will they finish it together?

Answer: A's rate = $1/12$, B's rate = $1/18$. Combined rate = $1/12 + 1/18 = 3/36 + 2/36 = 5/36$. Time together = $36/5 = 7.2$ days.

Tip: Work problems: always add rates (1/time), never add the times directly.

Q7. A sum of money doubles itself in 8 years at simple interest. Find the rate of interest.

Answer: Let Principal = P, so Interest earned = P in 8 years. Using $SI = (P \times R \times T)/100 \rightarrow P = (P \times R \times 8)/100 \rightarrow R = 100/8 = 12.5\%$.

Tip: For 'doubles in N years' at SI, $\text{Rate} = 100/N$.

Q8. What is the probability of getting exactly one head when two fair coins are tossed?

Answer: Sample space = {HH, HT, TH, TT}, total 4 outcomes. Favorable outcomes (exactly one head) = {HT, TH} = 2. Probability = $2/4 = 1/2$.

Tip: List the full sample space for small problems instead of using formulas blindly.

2. Logical & Analytical Reasoning

Tests pattern recognition and structured thinking — common filter round before the technical interview.

Q9. Find the next number in the series: 2, 6, 12, 20, 30, ?

Answer: Differences between terms: 4, 6, 8, 10, so next difference is 12. $30 + 12 = 42$. (Pattern: $n \times (n+1)$, i.e. $1 \times 2, 2 \times 3, 3 \times 4 \dots 6 \times 7 = 42$)

Tip: Check second-level differences when the first-level differences aren't constant.

Q10. If 'CODING' is written as 'DPEJOH', how is 'PYTHON' written in the same code?

Answer: Each letter is shifted forward by 1 (C→D, O→P, D→E ...). Applying the same shift to PYTHON: P→Q, Y→Z, T→U, H→I, O→P, N→O = QZUIPO.

Tip: For letter-coding questions, first find the shift pattern from the given example.

Q11. Statements: All data scientists are analysts. Some analysts are managers.

Conclusion: Some data scientists are managers. Is this conclusion valid?

Answer: Not necessarily valid. The two statements only guarantee an overlap between analysts and managers, and between data scientists and analysts — not a guaranteed overlap between data scientists and managers.

Tip: In syllogisms, draw simple Venn diagrams before deciding validity.

Q12. A is the father of B. B is the sister of C. C is the son of D. How is D related to A?

Answer: D is the spouse (wife) of A, since both A and D are parents of B and C.

Tip: Draw a quick family tree for relationship puzzles instead of tracking it mentally.

Q13. Pointing to a photograph, a man says, 'She is the daughter of my grandfather's only son.' How is the woman related to the man?

Answer: 'My grandfather's only son' = the man's father (assuming the man is that son). So the woman is the man's sister.

Tip: Watch for the phrase 'only son/daughter' — it pins down the relationship precisely.

Q14. Which number should replace the question mark? 7, 26, 63, 124, 215, ?

Answer: Each term = $n^3 - 1$, for $n = 2, 3, 4, 5, 6, 7$: $8-1=7$, $27-1=26$, $64-1=63$, $125-1=124$, $216-1=215$, so next = $7^3 - 1 = 343 - 1 = 342$.

Tip: For cubic/quadratic series, test n^3 or n^2 patterns against the given terms.

3. Data Interpretation

Reading and reasoning over tables/charts — directly relevant to a data scientist's daily work.

Q15. A company's revenue grew from ■40 lakh to ■58 lakh in one year. What is the percentage growth?

Answer: Growth = $(58 - 40)/40 \times 100 = 18/40 \times 100 = 45\%$.

Tip: Percentage change formula: $(\text{New} - \text{Old})/\text{Old} \times 100$.

Q16. A pie chart shows a company's expenses: Salaries 40%, Marketing 25%, R&D; 20%, Others 15%. If total expense is ■80 lakh, how much is spent on R&D;?

Answer: R&D; spend = 20% of 80 lakh = $0.20 \times 80 = \text{■}16$ lakh.

Tip: Convert pie chart percentages directly against the given total; no need to find other slices.

Q17. In a dataset, the mean of 10 values is 50. A data entry error is found: one value recorded as 40 should actually be 60. What is the corrected mean?

Answer: Original sum = $10 \times 50 = 500$. Corrected sum = $500 - 40 + 60 = 520$. Corrected mean = $520/10 = 52$.

Tip: For 'correction' problems, adjust only the total sum, then divide again by count.

Q18. A bar chart shows monthly active users (in thousands): Jan 20, Feb 25, Mar 22, Apr 30, May 28. Find the month-over-month growth rate from March to April.

Answer: Growth = $(30 - 22)/22 \times 100 = 8/22 \times 100 \approx 36.4\%$.

Tip: Always read chart values carefully before computing — DI errors are usually reading mistakes, not math mistakes.

4. Probability & Statistics

The most important section for a data science role — directly tests concepts used in ML and analytics.

Q19. A disease affects 1% of a population. A test for the disease is 99% accurate (both sensitivity and specificity). If a person tests positive, what is the probability they actually have the disease? (Bayes' Theorem)

Answer: Let D = has disease, T = tests positive. $P(D) = 0.01$, $P(T|D) = 0.99$, $P(T|\text{not } D) = 0.01$. $P(T) = P(T|D)P(D) + P(T|\text{not } D)P(\text{not } D) = (0.99 \times 0.01) + (0.01 \times 0.99) = 0.0099 + 0.0099 = 0.0198$. $P(D|T) = (0.99 \times 0.01) / 0.0198 = 0.0099 / 0.0198 = 0.5$, i.e. only 50%.

Tip: This classic result shows why base rate (prior probability) matters as much as test accuracy — a favorite DS interview question.

Q20. Differentiate between Type I and Type II errors in hypothesis testing.

Answer: Type I error (false positive) is rejecting a true null hypothesis — concluding there is an effect when there isn't one. Type II error (false negative) is failing to reject a false null hypothesis — missing a real effect. Type I is governed by significance level (alpha), Type II is related to statistical power ($1 - \beta$).

Tip: Remember: Type I = 'crying wolf', Type II = 'missing the wolf'.

Q21. What is the difference between correlation and causation? Give a short example.

Answer: Correlation means two variables move together statistically, but this does not imply one causes the other. Example: ice cream sales and drowning incidents both rise in summer — they are correlated through a hidden third variable (hot weather), not because one causes the other.

Tip: Interviewers love this question to test if a candidate blindly trusts statistical patterns.

Q22. In a normal distribution, approximately what percentage of data lies within 1, 2, and 3 standard deviations of the mean?

Answer: This is the Empirical (68-95-99.7) Rule: about 68% of data lies within ± 1 SD, about 95% within ± 2 SD, and about 99.7% within ± 3 SD of the mean.

Tip: Memorize 68-95-99.7 — it comes up constantly in outlier detection and A/B testing questions.

Q23. What is the difference between population variance and sample variance formulas?

Answer: Population variance divides the sum of squared deviations by N (total population size). Sample variance divides by $(n - 1)$ instead of n — this is called Bessel's correction, and it corrects the bias that occurs when estimating variance from a smaller sample.

Tip: If asked 'why n-1?', mention it keeps the sample variance an unbiased estimator of the population variance.

Q24. A bag contains 4 red and 6 blue balls. Two balls are drawn without replacement. What is the probability that both are red?

Answer: $P(\text{1st red}) = 4/10$. $P(\text{2nd red} | \text{1st red}) = 3/9$. $P(\text{both red}) = (4/10) \times (3/9) = 12/90 = 2/15$.

Tip: 'Without replacement' means the denominator shrinks by 1 for the second draw.

Q25. What is a p-value, and what does $p < 0.05$ mean in a hypothesis test?

Answer: The p-value is the probability of observing results at least as extreme as the current data, assuming the null hypothesis is true. $p < 0.05$ conventionally means there is less than a 5% chance of seeing this result by random chance alone, so the null hypothesis is rejected.

Tip: Common trap: a p-value is NOT the probability that the null hypothesis is true.

Q26. Explain the difference between mean, median, and mode, and when median is preferred over mean.

Answer: Mean is the arithmetic average, median is the middle value when data is sorted, and mode is the most frequent value. Median is preferred over mean when data has outliers or is skewed, since extreme values distort the mean but not the median.

Tip: Example to mention: median household income is used instead of mean because a few very high incomes would skew the mean upward.

5. SQL & Technical Aptitude

Basic technical-aptitude style questions that frequently appear alongside quant/reasoning rounds for data roles.

Q27. What is the difference between WHERE and HAVING clauses in SQL?

Answer: WHERE filters individual rows before any grouping/aggregation happens, and cannot use aggregate functions. HAVING filters groups after a GROUP BY, and is used specifically to filter based on aggregate values like COUNT() or SUM().

Tip: Quick line for interviews: 'WHERE filters rows, HAVING filters groups.'

Q28. Write the SQL query to find the second-highest salary from an Employee table.

Answer: `SELECT MAX(salary) FROM Employee WHERE salary < (SELECT MAX(salary) FROM Employee);` This finds the highest salary that is still less than the overall maximum.

Tip: Alternative: use DENSE_RANK() with a window function for the Nth-highest salary in general.

Q29. What is the difference between INNER JOIN and LEFT JOIN?

Answer: INNER JOIN returns only the rows that have matching values in both tables. LEFT JOIN returns all rows from the left table, along with matched rows from the right table — unmatched right-table columns are filled with NULL.

Tip: Draw two overlapping circles (Venn diagram) mentally to recall join behavior quickly.

Q30. What is the time complexity of binary search, and what precondition must the data satisfy?

Answer: Binary search runs in $O(\log n)$ time. The precondition is that the data (array/list) must already be sorted, since the algorithm repeatedly halves the search range based on comparisons.

Tip: Contrast with linear search at $O(n)$ — a common follow-up question.

Q31. In Python, what is the difference between a list and a tuple?

Answer: A list is mutable (can be changed after creation) and defined with square brackets []. A tuple is immutable (cannot be changed after creation) and defined with parentheses (). Tuples are generally faster and used for fixed collections of data.

Tip: Mention immutability as the reason tuples can be used as dictionary keys, while lists cannot.

Q32. What does the GROUP BY clause do, and how is it typically used with aggregate functions?

Answer: GROUP BY groups rows that share the same values in specified columns into summary rows, and is typically combined with aggregate functions like COUNT(), SUM(), AVG(), MIN(), or MAX() to compute a value per group, e.g. total sales per region.

Tip: Every non-aggregated column in the SELECT list must also appear in the GROUP BY clause.

How to Use This Guide

- Attempt each question yourself first before reading the answer — aptitude tests are timed, so practice recall speed.
- Re-derive the Bayes' theorem and probability answers by hand at least twice; these are the most commonly asked data-science-specific questions.
- For SQL questions, practice writing the query in an actual editor (e.g. SQLite/MySQL) rather than just reading it.
- Revisit the Quantitative and Reasoning sections a day before the test — they rely on speed more than deep understanding.

Good luck with your data science aptitude test and interview preparation!